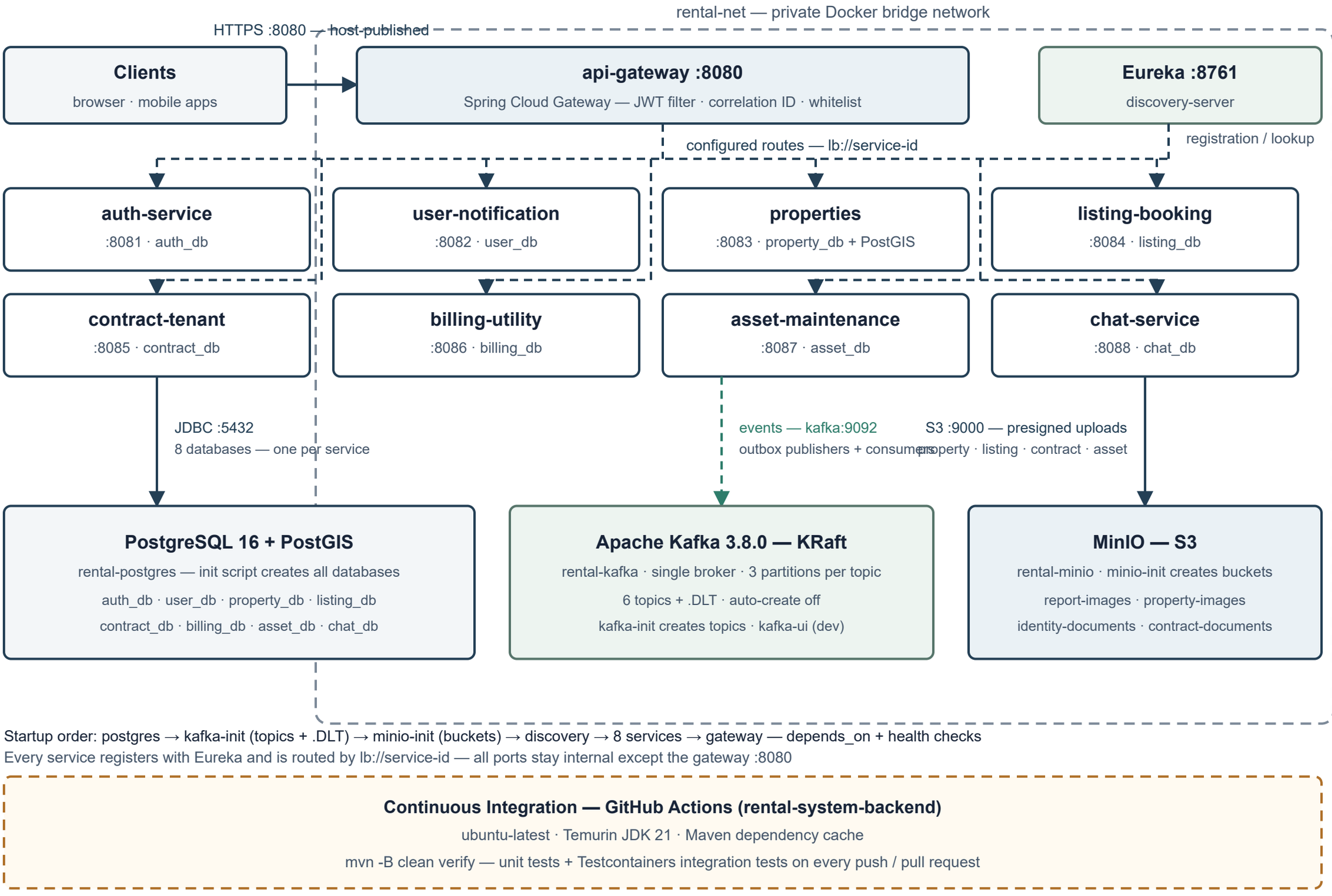

rental-net — private Docker bridge network
HTTPS :8080 — host-published
Clients
browser · mobile apps
api-gateway :8080
Spring Cloud Gateway — JWT filter · correlation ID · whitelist
Eureka :8761
discovery-server
registration / lookup
configured routes — lb://service-id
auth-service
:8081 · auth_db
user-notification
:8082 · user_db
properties
:8083 · property_db + PostGIS
listing-booking
:8084 · listing_db
contract-tenant
:8085 · contract_db
billing-utility
:8086 · billing_db
asset-maintenance
:8087 · asset_db
chat-service
:8088 · chat_db
JDBC :5432
8 databases — one per service
events — kafka:9092
outbox publishers + consumers
S3 :9000 — presigned uploads
property · listing · contract · asset
PostgreSQL 16 + PostGIS
rental-postgres — init script creates all databases
auth_db · user_db · property_db · listing_db
contract_db · billing_db · asset_db · chat_db
Apache Kafka 3.8.0 — KRaft
rental-kafka · single broker · 3 partitions per topic
6 topics + .DLT · auto-create off
kafka-init creates topics · kafka-ui (dev)
MinIO — S3
rental-minio · minio-init creates buckets
report-images · property-images
identity-documents · contract-documents
Startup order: postgres → kafka-init (topics + .DLT) → minio-init (buckets) → discovery → 8 services → gateway — depends_on + health checks
Every service registers with Eureka and is routed by lb://service-id — all ports stay internal except the gateway :8080
Continuous Integration — GitHub Actions (rental-system-backend)
ubuntu-latest · Temurin JDK 21 · Maven dependency cache
mvn -B clean verify — unit tests + Testcontainers integration tests on every push / pull request